ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
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17.1.2IO Basic Configuration Checklist

Table 17-3. Immediate Occupancy Basic Configuration Checklist.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Building System—General							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	LOAD PATH: The structure contains a complete, well-defined load path, including structural elements and connections, that serves to transfer the inertial forces associated with the mass of all elements of the building to the foundation.	5.4.1.1	A.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	ADJACENT BUILDINGS: The clear distance between the building being evaluated and any adjacent building is greater than 0.5% of the height of the shorter building in low seismicity, 1.0% in moderate seismicity, and 3.0% in high seismicity.	5.4.1.2	A.2.1.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	MEZZANINES: Interior mezzanine levels are braced independently from the main structure or are anchored to the seismic-force-resisting elements of the main structure.	5.4.1.3	A.2.1.3	
Building System—Building Configuration							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WEAK STORY: The sum of the shear strengths of the seismic-force-resisting system in any story in each direction is not less than 80% of the strength in the adjacent story above.	5.4.2.1	A.2.2.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SOFT STORY: The stiffness of the seismic-force-resisting system in any story is not less than 70% of the seismic-force-resisting system stiffness in an adjacent story above or less than 80% of the average seismic-force-resisting system stiffness of the three stories above.	5.4.2.2	A.2.2.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	VERTICAL IRREGULARITIES: All vertical elements in the seismic-force-resisting system are continuous to the foundation.	5.4.2.3	A.2.2.4	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
C	NC	N/A	U	GEOMETRY: There are no changes in the net horizontal dimension of the seismic-force-resisting system of more than 30% in a story relative to adjacent stories, excluding 1-story penthouses and mezzanines.	5.4.2.4	A.2.2.5	
C	NC	N/A	U	MASS: There is no change in effective mass of more than 50% from one story to the next. Light roofs, penthouses, and mezzanines need not be considered.	5.4.2.5	A.2.2.6	
C	NC	N/A	U	TORSION: The estimated distance between the story center of mass and the story center of rigidity is less than 20% of the building width in either plan dimension. This statement does not apply to buildings with flexible diaphragms.	5.4.2.6	A.2.2.7	
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
Geologic Site Hazards							
C	NC	N/A	U	LIQUEFACTION: Liquefaction-susceptible, saturated, loose granular soils that could jeopardize the building's seismic performance do not exist in the foundation soils at depths within 50 ft (15.2 m) under the building.	5.4.3.1	A.6.1.1	
C	NC	N/A	U	SLOPE FAILURE: The building site is located away from potential earthquake-induced slope failures or rockfalls so that it is unaffected by such failures or is capable of accommodating any predicted movements without failure.	5.4.3.1	A.6.1.2	
C	NC	N/A	U	SURFACE FAULT RUPTURE: Surface fault rupture and surface displacement at the building site are not anticipated.	5.4.3.1	A.6.1.3	
Tsunami Hazards							
C	NC	N/A	U	TSUNAMI: The building is not located within a Tsunami Design Zone as defined by ASCE 7 Chapter 6 or is located in a Tsunami Design Zone where the inundation depth per ASCE 7 Chapter 6 is less than 3 ft (0.9 m).	5.4.3.1	A.6.1.4	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate and High Seismicity (Complete the Following Items in Addition to the Items for Low Seismicity)							
Foundation Configuration							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TIES BETWEEN FOUNDATION ELEMENTS: For buildings supported on soils classified as Site Class D, DE, E, or F, the individual pile caps, piles, and piers are restrained by concrete beams or slabs adequate to resist seismic forces. For buildings supported on soils classified as Site Class E or F, individual spread footings are restrained by concrete beams or slabs adequate to resist seismic forces.	5.4.3.4	A.6.2.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DEEP FOUNDATIONS: Piles that are required to transfer lateral and/or overturning forces between the structure and the soil shall have a positive connection between the piles and the pile cap, foundation mat, grade beam, or other element of the building foundation system. Cast-in-place and precast non-prestressed piles shall have a minimum longitudinal reinforcement ratio of 0.0025 and transverse reinforcing spaced at no more than 6 in (152.4 mm) within a distance of three times the pile diameter from the bottom of the pile cap. Precast prestressesd piles shall have a minimum effective prestress of 400 psi and transverse reinforcing spaced at no more than 6 in. (152.4 mm) within a distance of 20 ft (6 m) from the top of the pile.		A.6.2.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SLOPING SITES: The exterior grade difference from one side of the building to another does not exceed one story in height.		A.6.2.4	

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